INCH-POUND
MIL-DTL-23765/1F(SH)
16 August 2024
SUPERSEDING
MIL-E-23765/1E(SH)
23 August 1985

DETAIL SPECIFICATION SHEET

ELECTRODES AND RODS – WELDING, BARE, SOLID AND ALLOY-CORED, ORDINARY STRENGTH AND LOW-ALLOY STEEL

This specification is approved for use by the Naval Sea System Command and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-23765.

1. SCOPE

1.1 Scope. This specification covers ordinary strength and low-alloy steel, bare, solid and alloy-cored, welding electrodes and cut-length rods for use with the gas metal-arc welding (GMAW), gas tungsten-arc welding (GTAW), and submerged-arc welding (SAW) processes. This specification also covers a neutral granular type flux for SAW.

1.2 Classification. Electrodes and rods are provided in the types specified in tables I and II and in forms 3a, 3b, 3c, 3d, 3e, 4, and 6 with sizes in accordance with MIL-DTL-23765 and table III herein.

1.2.1 Neutral granular flux type. Neutral granular flux for as-welded and stress-relieved SAW applications are provided in the types qualified in 3.3.3.

Comments, suggestions, or questions on this document should be addressed to Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard DC 20376-5160 or emailed to CommandStandards@navy.mil, with the subject line "Document Comment". Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

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TABLE I. Type designation and chemical composition for 70 series electrodes and rods.

MIL-type (with suffix "RC") 2/	Process	Chemical composition (percentage) 1/						
		Carbon	Manganese	Silicon	Phosphorus 2/	Sulfur	Copper	Other
70S-1	GMAW, GTAW, SAW	0.07–0.19	0.90–1.40	0.15–0.35	0.025	0.035	2/, 3/	----
70S-2	GMAW, GTAW, SAW	0.06	1.00–1.40	0.40–0.70	0.025	0.035		4/
70S-3 70S-3H	GMAW, GTAW, SAW	0.07–0.19	0.90–1.40	0.40–0.70	0.025	0.035		----
70S-4	GMAW, GTAW, SAW	0.07–0.15	0.90–1.40	0.65–0.85	0.025	0.035		----
70S-5	GMAW, GTAW, SAW	0.07–0.19	0.90–1.40	0.30–0.60	0.025	0.035		0.50–0.90 aluminum
70S-6	GMAW, GTAW, SAW	0.08–0.14	1.40–1.70	0.80–1.15	0.025	0.035		----
70S-7	SAW	0.10–0.18	1.70–2.10	0.05	0.025	0.035		----
70S-8	SAW	0.08–0.12	0.90–1.40	0.05	0.025	0.035		0.40–0.60 molybdenum
70S-9	SAW	0.10–0.18	1.70–2.10	0.05	0.025	0.035		0.40–0.60 molybdenum

NOTES:

1/ Wherever single values are shown, they are maximum values. Values apply to bare, solid electrode, rod, or weld deposit as specified in 3.4.

2/ Addition of the suffix "RC" to any basic MIL-type designation (e.g., MIL-70S-lRC) indicates a special MIL-type of electrode or rod that is not copper coated and for which the copper and phosphorus content are 0.40 percent and 0.012 percent maximum, respectively. Other requirements of this specification, which apply to a basic MIL-type, also apply to the special MIL-type counterpart with the restricted copper and phosphorus content (see 6.2).

3/ When the basic MIL-type electrode or rod has a copper coating in accordance with 3.3.1 herein, the maximum weight percent of copper in the electrode or rod due to the coating and the residual copper content in the steel is 0.30 percent maximum.

4/ Aluminum 0.05 to 0.15, zirconium 0.02 to 0.12, and titanium 0.05 to 0.15.

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TABLE II. Type designation and chemical composition for 80 series electrodes and rods.

Element [1/]	MIL-type [2/, 3/]		
	80S-1	**80S-2**	**80S-3**
Carbon	0.10–0.15	0.10–0.17	0.07–0.12
Manganese	1.75–2.40	1.70–2.20	1.60–2.10
Silicon	0.10	0.05	0.50–0.80
Phosphorus	0.025	0.025	0.025
Sulfur	0.020	0.020	0.020
Nickel	0.55–1.10	0.15	0.15
Molybdenum	0.35–0.55	0.45–0.60	0.40–0.60
Chromium	0.15	0.15	0.15
Vanadium	0.03	0.03	0.03
Aluminum	0.07	0.07	0.02
Copper	[4/, 5/]	[4/, 5/]	[4/, 5/]

NOTES:

[1/] The units for numerical values shown for each element are in percentage. Wherever single values are shown, they are maximum values. Values apply to bare, solid electrode, rod, or weld deposit as specified in 3.4.

[2/] Including applicable suffixes (see 3.3).

[3/] The applicable welding process for MIL-80S-1 and 80S-2 is SAW. Applicable welding processes for MIL-80S-3 are GMAW and GTAW.

[4/] When the basic MIL-type electrode or rod has a copper coating in accordance with 3.3.1 herein, the maximum weight percent of copper in the electrode or rod due to the coating and the residual copper content in the steel is 0.30 percent maximum.

[5/] Addition of the suffix “RC” to any basic MIL-type designation (e.g., MIL-80S-1RC) indicates a special MIL-type of electrode or rod that is not copper coated and for which the copper is 0.10 percent maximum. Other requirements of this specification that apply to a basic MIL-type also applies to the special MIL-type counterpart with the restricted copper content (see 6.2).

TABLE III. Form, size, and weight.

Form	Electrode diameter (inch)	Weight (pounds) [1/]
3a	All	1½ and 2½
3b	All	10, 12, 15, 25, 30, 35, 44, 50, or 60
3c	All	25, 50, or 60
3d	All	60 or 65
3e	All	150, 200, or 250
4	All	As specified by the purchaser
6	All	10 or 50

NOTE:

[1/] Tolerance on net weight is ±10 percent.

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2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-S-22698	-	Steel Plate, Shapes and Bars, Weldable Ordinary Strength and Higher Strength: Structural
MIL-DTL-23194	-	Steel, Forgings, Carbon and Low Alloy (UNS K13502, K12766, and K42365)
MIL-DTL-23765	-	Electrodes and Rods – Welding, Bare, Solid and Alloy-Cored, General Specification for
MIL-S-24238	-	Steel Plate, Carbon and Low Alloy

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-2035	-	Nondestructive Testing Acceptance Criteria

(Copies of these documents are available online at https://quicksearch.dla.mil.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

NAVAL SEA SYSTEMS COMMAND (NAVSEA) PUBLICATIONS

S9074-AR-GIB-010/278	-	Requirements for Fabrication Welding and Inspection, and Casting Inspection and Repair for Machinery, Piping, and Pressure Vessels

(Copies of this document are available online via Model Based Product Support (MBPS) at https://mbps.navseaplm.navy.mil/Windchill/app/. To gain access to MBPS, obtain an account with National Help Desk Service Management (NHDSM) at https://nhdsm.navair.navy.mil (a valid CAC is required to access this website) and submit a SAAR-N Request. Refer questions, inquiries, or problems to (888) 292-5919. This document is available for ordering (hard copy) via the Naval Logistics Library (NLL) at https://nll.navsup.navy.mil. For questions regarding the NLL, contact the NLL Customer Service at nllhelpdesk@navy.mil, (866) 817-3130, or (215) 697-2626/DSN 442-2626.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI/AWS A4.3	-	Standard Methods for Determination of the Diffusible Hydrogen Content of Martensitic, Bainitic, and Ferritic Steel Weld Metal Produced by Arc Welding

(Copies of this document are available online at https://webstore.ansi.org/.)

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AMERICAN WELDING SOCIETY (AWS)

AWS A3.0M/A3.0	-	Standard Welding Terms and Definitions; Including Terms for Adhesive Bonding, Brazing, Soldering, Thermal Cutting, and Thermal Spraying
AWS A5.32M/A5.32	-	Welding Consumables – Gases and Gas Mixtures for Fusion Welding and Allied Processes

(Copies of these document are available online at www.aws.org.)

ASTM INTERNATIONAL

ASTM A285/A285M	-	Standard Specification for Pressure Vessel Plates, Carbon Steel, Low- and Intermediate-Tensile Strength
ASTM A302/A302M	-	Standard Specification for Pressure Vessel Plates, Alloy Steel, Manganese-Molybdenum and Manganese-Molybdenum-Nickel
ASTM A508/A508M	-	Standard Specification for Quenched and Tempered Vacuum-Treated Carbon and Alloy Steel Forgings for Pressure Vessels
ASTM A516/A516M	-	Standard Specification for Pressure Vessel Plates, Carbon Steel, for Moderate- and Lower-Temperature Service
ASTM A533/A533M	-	Standard Specification for Pressure Vessel Plates, Alloy Steel, Quenched and Tempered, Manganese-Molybdenum and Manganese-Molybdenum-Nickel
ASTM A945/A945M	-	Standard Specification for High-Strength Low-Alloy Structural Steel Plate with Low Carbon and Restricted Sulfur for Improved Weldability, Formability, and Toughness
ASTM E350	-	Standard Test Methods for Chemical Analysis of Carbon Steel, Low-Alloy Steel, Silicon Electrical Steel, Ingot Iron, and Wrought Iron

(Copies of these documents are available online at www.astm.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 General. Electrodes, rods, and neutral granular flux furnished under this specification shall be in accordance with MIL-DTL-23765 and as specified herein.

3.2 Qualification. Electrodes, rods, and neutral granular flux furnished under this specification shall be products that are authorized by the qualifying activity for listing on the applicable qualified products list (QPL) before contract award (see 4.2 and 6.3).

3.3 MIL-types.

3.3.1 Basic MIL-type. Basic MIL-type electrodes and rods may have either a clean, bright finish or a uniform continuous well-bonded smoothly drawn copper coating on a clean surface. Diameters 3/32 inch and smaller may be coated with other types of rust preventatives, provided such coatings do not impair usability of the electrodes or rods or the quality or soundness of weld metal deposits.

3.3.2 Special MIL-type with suffix "RC". Special MIL-type electrodes and rods, designated by the suffix "RC" , shall not be copper coated but shall have either a bright finish or a special protective coating, provided the coating does not impair usability of the electrodes or rods or the quality or soundness of the weld metal deposits.

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3.3.3 MIL-type with suffix “F”. MIL-70S-F and MIL-80S-F are generic designators for fluxes. The specific MIL-type designation of a qualified flux consists of the MIL-type designation of the electrode it was qualified with, plus the suffix “F” (e.g., MIL-70S-1F or MIL-80S-2F).

3.3.3.1 Wall neutrality number (N). Neutral granular flux, type MIL-70S-F, shall have a Wall neutrality number (N) of 35 or less.

3.3.4 MIL-type with suffix “C”. Alloy-cored electrodes shall be designated by suffix “C” (e.g., MIL-70S-1C).

3.4 Chemical composition. The chemical composition of unwelded electrodes, rods, and weld deposits shall be in accordance with table I (MIL-70 series) and table II (MIL-80 series).

3.5 Mechanical properties. Unless otherwise specified (see 6.2), the mechanical properties for as-welded weld metal shall be as specified in table IV (MIL-70S series). The mechanical properties for stress-relieved weld metal deposited by the SAW, GMAW, and GTAW processes shall be as specified in table V (MIL-70S series) and table VI (MIL-70S and 80S series).

TABLE IV. Weld metal mechanical properties deposited with GMAW, GTAW, and SAW for 70 series electrodes, rods, and fluxes (as-welded condition). 1/

MIL-type 2/	Ultimate tensile strength (minimum pounds per square inch [psi])	Yield strength at 0.2 percent offset (psi)	Percent elongation, in 2 inches	Bend	Charpy V-notch (CVN) impact test temperature (°F)	Average CVN impact energy, (minimum foot-pound [ft-lb])
70S-1	70,000	55,000 to 80,000 3/	22	4/	minus 20	20
70S-2					minus 20	20
70S-3					minus 20	20
70S-4					minus 20	20
70S-5					0	20
70S-6					minus 20	20
70S-3H		55,000 to 90,000 3/			minus 20	30

NOTES:

1/ Single values are minimum.

2/ Including applicable suffixes (see 3.3).

3/ For GTAW, the yield strength shall be 55,000 psi minimum.

4/ The specimens shall have no cracks on the convex surface. Tears less than ⅛ inch on the corner of the bend specimen are acceptable.

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TABLE V. Weld metal properties for stress-relieved specimens cooled at 200 °F per hour (see 3.5.1 and 3.5.1.1).

MIL-type [1/]	70S-1, -2, -3, -4, and-6 [2/]	70S-5
Ultimate tensile strength, minimum psi	70,000	70,000
Yield strength at 0.2 percent offset, minimum psi	52,000	52,000
Percent elongation in 2 inches, minimum	26	26
CVN impact test temperature, °F [2/]	minus 20	0
Average CVN impact energy, minimum ft-lb [3/]	20	20

NOTES:

1/ Including applicable suffixes (see 3.3).

2/ Impact properties for the five specimens for each test condition shall be obtained at test temperatures as specified ±3 °F.

3/ Individual impact results shall be recorded. At the temperature specified, the average value of five tests shall be equal to or greater than the value specified. The results used in determining each average shall contain no more than one observation below the specified value and that observation shall be not more than 5 ft-lb below the specified value.

TABLE VI. Weld metal properties for stress-relieved specimens cooled at 10 °F per hour (see 3.5.1 and 3.5.1.2).

MIL-type [1/]	70S-1, -2, -3, -4, and -6	70S-5	70S-7	70S-8	70S-9	80S-1, -2, and -3
Ultimate tensile strength, minimum psi	70,000	70,000	70,000	70,000	70,000	80,000
Yield strength at 0.2 percent offset, minimum psi	38,000	38,000	38,000	38,000	38,000	50,000
Percent elongation in 2 inches, minimum	26	26	22	22	22	20
CVN impact test temperature, °F [2/]	minus 20	0	plus 10	plus 10	plus 10	plus 10
Average CVN impact energy, minimum ft-lb [3/]	20	20	15	25	30	30
Transverse side bend	----	----	----	----	----	[4/]

NOTES:

1/ Including applicable suffixes (see 3.3).

2/ Impact properties for the five specimens for each test condition shall be obtained at test temperatures as specified ±3 °F.

3/ Individual impact results shall be recorded. At the temperature specified, the average value of five tests shall be equal to or greater than the value specified. The results used in determining each average shall contain no more than one observation below the specified value and that observation shall be not more than 5 ft-lb below the specified value.

4/ The specimens shall have no cracks on the convex surface. Tears less than ⅛ inch on the corner of the bend specimen are acceptable.

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3.5.1 Stress relief. General requirements for stress relief shall be in accordance with S9074-AR-GIB-010/278. Holding temperatures, time at holding temperature, and cooling rates shall be as specified in 3.5.1.1 or 3.5.1.2, as applicable (see tables VIII, X, XII, and XIV).

3.5.1.1 Stress relief (200 Fahrenheit [°F] per hour cooling rate). Weldments shall be heated to 1125±25 °F, held at this temperature for 2 hours minimum, and furnace cooled at 200 °F per hour or slower to below 500 °F.

3.5.1.2 Stress relief (10 °F per hour cooling rate). Weldments shall be heated to 1125±25 °F for one of the following time periods: (a) 50 hours minimum or greater, or (b) held at this temperature for 1 hour per inch of weld thickness (for weld thicknesses less than 1 inch, the minimum holding time shall be proportional to the weld thickness but shall be not less than 30 minutes). The weldment treated by either method (a) or (b) shall be cooled at 10 °F per hour or slower to below 600 °F.

3.6 Slag removal. The slag deposited during SAW shall be capable of being readily removed from weld deposits with hand tools (not air or power operated).

3.7 Weld metal diffusible hydrogen. When diffusible hydrogen testing is specified herein, it shall not exceed an average value of 8 milliliter /100 gram when tested in accordance with AWS A4.3.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Conformance inspection (see 4.3).

4.2 Qualification inspection. Qualification inspection shall include the examinations of MIL-DTL-23765 and the tests specified in table VII.

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TABLE VII. Tests required for qualification inspection.

MIL-types [1/]	Form	Test required for qualification inspection											Test procedure	Requirements
		Cast	Helix	Chemical	Wall #	RT [2/]	Tensile [3/]	Bend	Impact	Alloy identity	Welded test assembly	Diffusible hydrogen		
All electrodes and rods	3b or 3d	X	X	----	----	----	----	----	----	----	----	----	MIL-DTL-23765	MIL-DTL-23765
		----	----	----	----	X	----	----	----	----	Table VIII (MIL-70 series) Table IX (MIL-70S-3H) Table X (MIL-80 series)	----	MIL-DTL-23765	MIL-DTL-23765
70S-1 70S-2 70S-3 70S-4 70S-5 70S-6	3b or 3d	----	----	X	----	----	----	----	----	----	----	----	3.4 herein and MIL-DTL-23765	Table I herein
		----	----	----	----	----	X	X	X	----	Table VIII	----	MIL-DTL-23765	For the as-welded condition: Table IV herein
		----	----	----	----	----	X	----	X	----	Table VIII	----	MIL-DTL-23765	For the stress-relieved condition: Tables V and VIII (note 3) herein
70S-3H	3b or 3d			X									MIL-DTL-23765	Table I herein
		----	----	----	----	----	X	X	X	----	Table IX	----	MIL-DTL-23765	For the as-welded condition: Table IV herein
		----	----	----	----	----	----	----	----	----	----	X [4/]	3.7 herein	3.7 herein
70S-7 70S-8 70S-9	3d	----	----	X	----	----	----	----	----	----	----	----	3.4 herein and MIL-DTL-23765	Table I herein
		----	----	----	----	----	X	----	X	----	Table VIII	----	MIL-DTL-23765	For the stress-relieved condition: Tables VI and VIII (note 3) herein

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TABLE VII. Tests required for qualification inspection – Continued.

MIL-types [1/]	Form	Test required for qualification inspection											Test procedure	Requirements
		Cast	Helix	Chemical	Wall #	RT [2/]	Tensile [3/]	Bend	Impact	Alloy identity	Welded test assembly	Diffusible hydrogen		
70S-F	3d	----	----	----	X		----	----	----	----	----	[5/]	4.2.1 herein	3.3.3.1 herein
		----	----	----	----	X	----	----	----	----	Table VIII	----	MIL-DTL-23765	MIL-DTL-23765
		----	----	----	----	----	X	X	X	----	Table VIII	----	MIL-DTL-23765	For the as-welded condition: Table IV herein
														For the stress-relieved condition: Tables V and VIII (note 3) herein
80S-1 80S-2 80S-3	3b or 3d	----	----	X	----	----	----	----	----	----	----	----	3.4 herein and MIL-DTL-23765	Table II herein
		----	----	----	----	----	----	----	----	X	----	----	MIL-DTL-23765	MIL-DTL-23765
		----	----	----	----	----	X	X	X		Table X	----	MIL-DTL-23765	For the stress-relieved condition: Tables VI and X (note 2) herein
80S-1F 80S-2F	3d	----	----	----	----	X	----	----	----	----	Table X	----	MIL-DTL-23765	MIL-DTL-23765
		----	----	----	----	----	X	X	X	----	Table X	----	MIL-DTL-23765	For the stress-relieved condition: Tables VI and X (note 2) herein

NOTES:

[1/] Includes applicable suffixes (see 3.3).

[2/] Radiographic Testing (RT).

[3/] Includes ultimate tensile strength, 0.2 percent offset yield strength, and percent elongation in 2 inches.

[4/] Diffusible hydrogen testing is only applicable to SAW electrodes and flux.

[5/] MIL-70S-3HF flux requires diffusible hydrogen testing in accordance with 3.7 herein.

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TABLE VIII. Qualification requirements for test assemblies using MIL-70 series (except for MIL-70S-3H) electrodes and fluxes. 1/, 2/

Purpose	Qualification	
Conditions (MIL-70S-1, -2, -3, -4, -5, and -6)	As-welded and stress-relieved 3/	
Condition (MIL-70S-7, -8, and -9)	Stress-relieved only 3/	
Welding parameters 4/, 5/	----	----
Welding process (MIL-70S-1, -2, -3, -4, -5, and -6)	SAW	GMAW
Welding process (MIL-70S-7, -8, and -9)	SAW	----
Position	1G 6/	1G 6/
Electrode size (inch) 7/	1/8	1/16
Polarity	DCEP 8/	DCEP 8/
Flux/shielding gas	70S-F 9/	10/
Joint configuration (as-welded and stress-relieved)	figure 1 11/, 12/	figure 1 11/, 12/
Base material	13/	13/
Thickness (T) (inch)	3/4	3/4
Width (W) (inches) (approximate)	10	10
Length (L) (inches) (minimum)	12	12
Root opening (R) (inch)	1/2	1/2
Root face (F) (inch) (maximum)	1/16	1/16
Included bevel angle (θ) (degrees)	45	45
Backing strap dimensions (t by w) (inches)	1/4 minimum thickness by 1	1/4 minimum thickness by 1
Test specimens (number/type) (as-welded) 14/	1 Tensile 2 Bend 5 CVN	1 Tensile 2 Bend 5 CVN
Test specimens (number/type) (stress-relieved) 14/	1 Tensile 5 CVN	1 Tensile 5 CVN

NOTES:

1/ Including applicable suffixes (see 3.3).

2/ For Wall neutrality number (N) requirements, see 4.2.1.

3/ For MIL-70S-1, -2, -3, -4, -5, and -6 electrodes, the stress relief shall be in accordance with 3.5.1 and 3.5.1.1. For MIL-70S-7, -8, and -9 electrodes, the stress relief shall be in accordance with 3.5.1 and 3.5.1.2 including methods (a) and (b).

4/ The manufacturer's procedure shall reflect all essential elements necessary for welding the test assembly.

5/ Welding currents and pass sequence shall be in accordance with sound welding practices and as recommended by the manufacturer.

6/ 1G is the welding position (flat) as defined by AWS A3.0M/A3.0.

7/ The sizes specified are used for qualification tests to qualify electrode sizes under 3/32 inch for the GMAW process or electrode sizes 3/32 inch and larger for the SAW process. Qualification of a solid wire electrode type for GMAW also qualifies it in the same size range for rods intended for use with GTAW.

8/ Direct Current Electrode Positive (DCEP).

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TABLE VIII. Qualification requirements for test assemblies using MIL-70 series (except for MIL-70S-3H) electrodes and fluxes – Continued. 1/, 2/

NOTES (continued):

9/ When qualifying MIL-type 70S electrodes, a qualified MIL-70 series flux of the corresponding type (e.g., MIL-70S-1 with MIL-70S-1F) in accordance with this specification shall be used. When qualifying MIL-70S-F fluxes, the flux shall be used in combination with ⅛-inch size electrode of the corresponding type (e.g., MIL-70S-1F with MIL-70S-1) to deposit weld metal by the SAW process. An electrode/flux combination may also be qualified simultaneously when tested together and meets all requirements for both the electrode and flux.

10/ Shielding gases shall be in accordance with AWS A5.32M/A5.32. Shielding gas for the GMAW process shall be as follows:

a. MIL-70S-1: Argon plus 2 percent oxygen
b. MIL-70S-2, -3, -4, and -6: One weld joint using argon plus 2 percent oxygen and one weld joint using carbon dioxide
c. MIL-70S-5: Carbon dioxide

11/ The assembly shall be at room temperature at start of welding, but not less than the required minimum preheat temperature. Continuous welding shall be employed in order to achieve the required interpass temperature. The interpass temperature range shall be maintained until welding is completed.

12/ After completion of each pass, the interpass temperature of the assembly shall be checked and the assembly shall be allowed to cool or shall be reheated, as required to maintain the assembly temperature in the required range prior to the resumption of welding. Any required cooling of test assemblies shall be done in still, room temperature air and assemblies shall not be force cooled with compressed air or by any other means.

13/ For each MIL-type, the base metal steel used for the tests required herein shall be in accordance with one of the following or equivalent:

a. Grade B steel of MIL-S-22698
b. Composition C of MIL-DTL-23194 or MIL-S-24238
c. Grade 70 of ASTM A516/A516M
d. Grade C of ASTM A285/A285M

14/ The tensile specimen shall be 0.500 inch in diameter in accordance with MIL-DTL-23765.

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TABLE IX. Qualification requirements for test assemblies using MIL-70S-3H series electrodes and flux. [1/, 2/]

Purpose	Qualification			
Condition (MIL-70S-3H)	As-welded only			
Welding parameters [3/, 4/]	----		----	
Welding process (MIL-70S-3H)	SAW		GMAW	
Electrode size (inch) [5/]	1/8		1/16	
Polarity	DCEP [6/]		DCEP [6/]	
Flux/shielding gas	70S-3HF [7/]		[8/]	
Cooling rate	High	Low	High	Low
Heat input (kilojoule/inch [kJ/in]) [9/]	30+2/-5	75+2/-5	30+2/-5	75+2/-5
Minimum preheat/maximum interpass temperature (°F) [10/]	60/150	275/300	60/150	275/300
Position	1G [11/]	1G [11/]	1G [11/]	3G [12/]
Joint configuration	figure 1		figure 1	
Base material	[13/]		[13/]	
Thickness (T) (inch)	3/4		3/4	
Width (W) (inches) (approximate)	10		10	
Length (L) (inches) (minimum)	12		12	
Root opening (R) (inch)	1/2		1/2	
Root face (F) (inch) (maximum)	1/16		1/16	
Included bevel angle (θ) (degrees)	45		45	
Backing strap dimensions (t by w) (inches)	1/4 minimum thickness by 1		1/4 minimum thickness by 1	
Test specimens (number/type) (as-welded) [14/]	1 Tensile 2 Bend 5 CVN		1 Tensile 2 Bend 5 CVN	

NOTES:

1/ Including applicable suffixes (see 3.3).

2/ For Wall neutrality number (N) requirements, see 4.2.1.

3/ The manufacturer's procedure shall reflect all essential elements necessary for welding the test assembly.

4/ Welding currents and pass sequence shall be in accordance with sound welding practices and as recommended by the manufacturer.

5/ The sizes specified are used for qualification tests to qualify electrode sizes under 3/32 inch for the GMAW process or electrode sizes 3/32 inch and larger for the SAW process. Qualification of a solid wire electrode type for GMAW also qualifies it in the same size range for rods intended for use with GTAW.

6/ Direct Current Electrode Positive (DCEP).

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TABLE IX. Qualification requirements for test assemblies using MIL-70S-3H series electrodes and flux – Continued. 1/, 2/

NOTES (continued):

7/ When qualifying MIL-type 70S-3H electrodes, a qualified MIL-70S-3HF flux in accordance with this specification shall be used. When qualifying MIL-70S-3HF flux, the flux shall be used in combination with ⅛-inch size electrode to deposit weld metal by the SAW process. An electrode/flux combination may also be qualified simultaneously when tested together and meets all requirements for both the electrode and flux.

8/ Shielding gases shall be in accordance with AWS A5.32M/A5.32. Shielding gas for the GMAW process shall be as follows:

a. MIL-70S-1: Argon plus 2 percent oxygen
b. MIL-70S-2, -3, -4, and -6: One weld joint using argon plus 2 percent oxygen and one weld joint using carbon dioxide
c. MIL-70S-5: Carbon dioxide

9/ The listed heat input shall be the average heat input considering all-weld passes. The root layer may be excluded from all heat input calculations.

10/ After the joint has been tacked, the assembly shall be preheated as required. After completion of each pass, the interpass temperature of the assembly shall be checked and the assembly shall be allowed to cool or shall be reheated, as required to maintain the assembly temperature in the required range prior to the resumption of welding. Any required cooling of test assemblies shall be done in still room temperature air and assemblies shall not be force cooled with compressed air or by any other means.

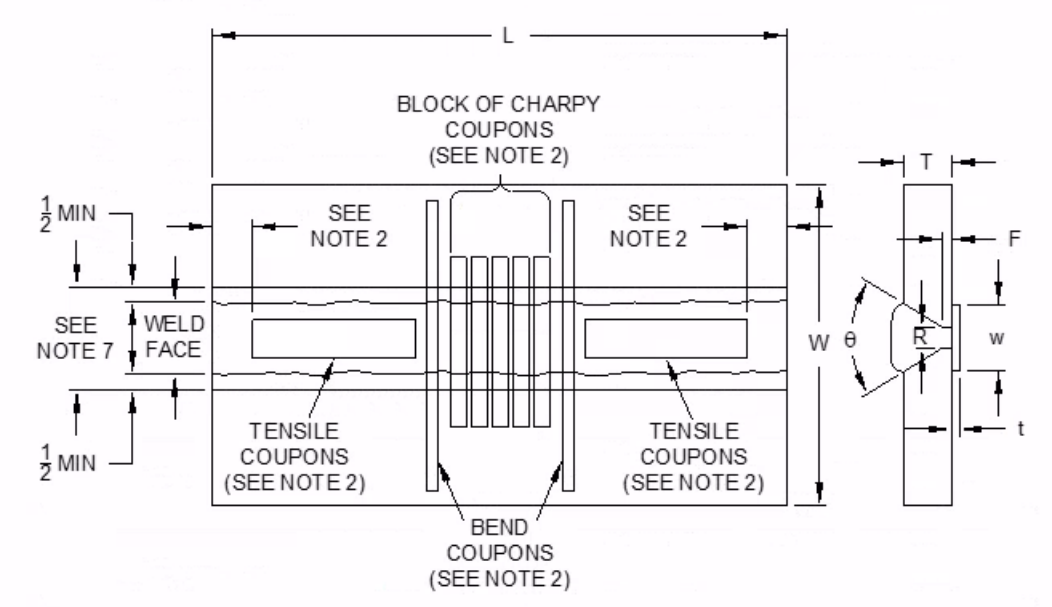
11/ 1G is the welding position (flat) as defined by AWS A3.0M/A3.0.

12/ 3G is the welding position (vertical) as defined by AWS A3.0M/A3.0.

13/ The base metal steel used for the test required herein shall be in accordance with ASTM A945/A945M, grade 65.

14/ The tensile specimen shall be 0.500 inch in diameter in accordance with MIL-DTL-23765.

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NOTES:

1. The as-deposited weld reinforcement shall be not greater than 3/16 inch for either the as-welded or the stress-relieved condition.
2. Specimen positioning along the length dimension is not mandatory, except as follows:
   a. The center section of the welded joint shall be used for the preparation of CVN impact specimens.
   b. No specimen shall be removed within ¾ inch of the end of the test assembly.
3. After completion of welding for the as-welded condition, the weld reinforcement and backing strap shall be removed flush with the base plate on both surfaces and the assembly shall be subjected to the additional required nondestructive inspections.
4. When testing in the stress-relieved condition is required, the weld assembly shall be stress-relieved prior to machining specimens and removing the backing strap. Subsequent to the stress-relief heat treatment, the weld assembly shall be allowed to cool and the weld reinforcement and backing straps shall be removed flush with the base plate on both surfaces and the assembly subjected to the additional required nondestructive inspections.
5. In preparation for performing the required nondestructive inspections, the following may be performed:
   a. Weld spatter, arc strikes, grinding marks, surface roughness, and handling marks that may interfere with inspection may be removed by wire brushing or grinding prior to radiography.
   b. Weld pickup (that is, welding performed prior to formal nondestructive inspection to correct minor visual surface irregularities, such as low areas in the weld reinforcement, weld bead roll, and a start-stop area) and minor repair (e.g., shallow lack of fusion against backing bar) may be accomplished on the weld face (before or after flush grinding or machining the weld reinforcement) and on the backside after backing bar removal provided that the same electrode being tested is used in conjunction with the same welding conditions, welding position, and other controls used in the original welding of the test assembly.

FIGURE 1. Joint design and specimen layout for as-welded and stress-relieved test assemblies.

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NOTES (continued):

6. Carbon arc or other thermal gouging operations shall not be performed on completed welds. This does not apply to plate preparations for the original joint.
7. Thermal cutting may be performed on completed welds provided cutting is not within ½ inch of the weld face.
8. Only sawing, machining, or both shall be used for removal of test coupons.
9. The centerline of the all-weld metal tensile specimens shall be nominally ⅜ inch below the top surface of the test plate.
10. The CVN specimens shall be removed and machined in accordance with MIL-DTL-23765 and to the details shown on figure 2.
11. Bend specimens (when required) shall be machined in accordance with MIL-DTL-23765.
12. For each test plate, a record of actual weld bead placement, actual root opening, and weld bead sequence shall be kept. The total number of weld passes, the actual root opening, and the actual bead placement shall be recorded with a schematic sketch or the use of written format for bead placement.

FIGURE 1. Joint design and specimen layout for as-welded and stress-relieved test assemblies – Continued.

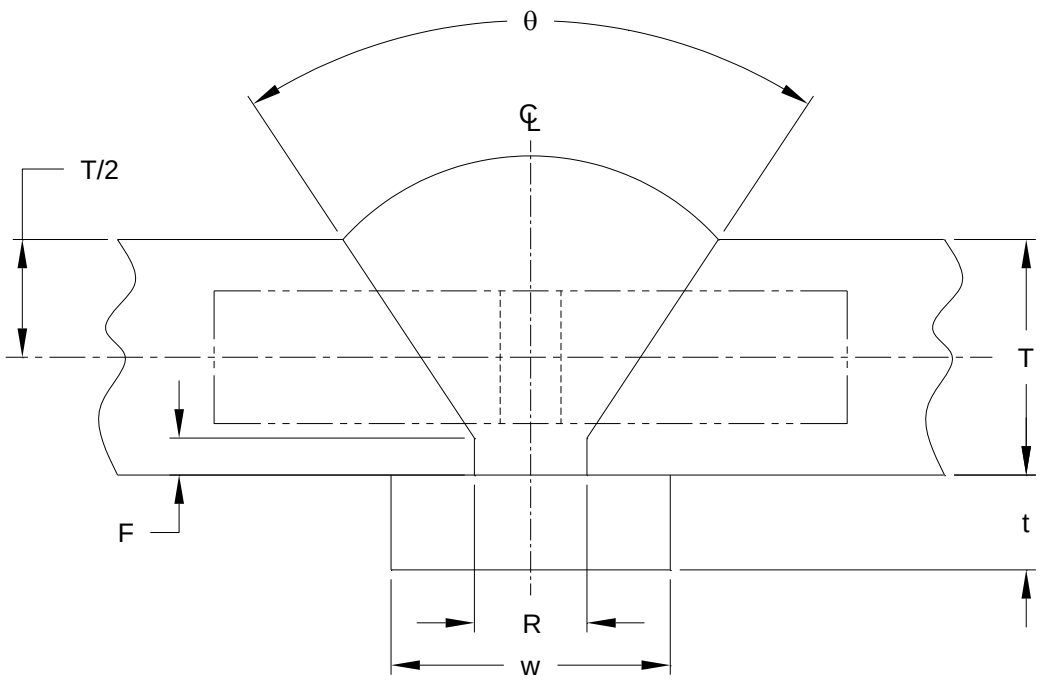


FIGURE 2. Orientation and positioning of CVN impact specimens.

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TABLE X. Qualification requirements for stress-relieved test assemblies for MIL-80 series electrodes and fluxes. [1/]

Purpose	Qualification	
Conditions (MIL-80S-1, -2, and -3)	Stress-relieved [2/]	
Welding parameters [3/, 4/]	----	----
Welding process (MIL-80S-1 and -2)	SAW	----
Welding process (MIL-80S-3)	----	GMAW (spray)
Position	1G [5/]	1G [5/]
Electrode size (inch) [6/]	1/8	1/16
Polarity	DCEP [7/]	DCEP [7/]
Flux/shielding gas	[8/]	[9/]
Minimum preheat/maximum interpass temperature (°F) [10/]	250/500	250/500
Joint configuration	figure 1	figure 1
Base material	[11/]	[11/]
Thickness (T) (inch)	3/4	3/4
Width (W) (inches) (approximate)	10	10
Length (L) (inches) (minimum)	18	18
Root opening (R) (inch)	1/2	1/2
Root face (F) (inch) (maximum)	1/16	1/16
Included bevel angle (θ) (degrees)	45	45
Backing strap dimensions (t by w) (inches)	1/4 to 3/8 by 1	1/4 to 3/8 by 1
Test specimens (number/type) [12/]	2 Tensile 2 Bend 5 CVN	2 Tensile 2 Bend 5 CVN

NOTES:

1/ Including applicable suffixes (see 3.3).

2/ Weldments shall be tested in the stress-relieved condition. The stress relief shall be in accordance with 3.5.1 and 3.5.1.2 including both methods (a) and (b).

3/ The manufacturer's procedure shall reflect all essential elements necessary for welding the test assembly.

4/ The welding currents and pass sequence shall be in accordance with sound welding practices and as recommended by the manufacturer.

5/ 1G is the welding position (flat) as defined by AWS A3.0M/A3.0.

6/ The sizes specified are used for qualification tests to qualify electrode sizes under 3/32 inch for the GMAW process or electrode sizes 3/32 inch and larger for the SAW process. Qualification of a solid wire electrode type for GMAW also qualifies it in the same size range for rods intended for use with GTAW.

7/ Direct Current Electrode Positive (DCEP).

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TABLE X. Qualification requirements for stress-relieved test assemblies for MIL-80 series electrodes and fluxes – Continued. 1/

NOTES (continued):

8/ When qualifying MIL-type 80S electrodes, a qualified MIL-80 series flux of the corresponding type (e.g., MIL-80S-1 with MIL-80S-1F) in accordance with this specification shall be used. When qualifying MIL-type 80S fluxes, the flux shall be used in combination with a ⅛-inch size electrode of the corresponding type (e.g., MIL-80S-1 with MIL-80S-1F) to deposit weld metal by the SAW process. An electrode and flux combination may also be qualified simultaneously when tested together and meets all requirements for both the electrode and flux.

9/ Shielding gases shall be in accordance with AWS A5.32M/A5.32. Shielding gas shall be argon plus 2 percent oxygen and argon plus 5 percent CO_2 for GMAW welded electrodes.

10/ After the joint has been tacked, the assembly shall be preheated. After completion of each pass, the interpass temperature of the assembly shall be checked and the assembly shall be allowed to cool or shall be reheated, as required to maintain the assembly temperature in the required temperature range prior to resumption of welding. Any required cooling of test assemblies shall be done in still, room temperature air and assemblies shall not be force cooled with compressed air or by any other means.

11/ For each MIL-type, the base metal steel used for the tests required herein shall be in accordance with one of the following:

a. Ni-Cr-Mo steel in accordance with MIL-DTL-23194, composition A
b. ASTM A508/A508M, grade 2, class 1 or equivalent
c. Mn-Ni-Mo steel in accordance with MIL-S-24238, composition A
d. ASTM A302/A302M
e. ASTM A533/A533M, type B or C, class 1, or equivalent

12/ The tensile specimen shall be 0.500 inch in diameter in accordance with MIL-DTL-23765.

4.2.1 Wall neutrality number (N). Prior to welding the test plates for flux qualification, the Wall neutrality number (N) for MIL-70S-F shall be determined in accordance with 4.2.1.1.

4.2.1.1 Determination of N. Determination of N shall be in accordance with the following:

a. Weld test pad (minimum size – 1 inch wide by 4 inches long by ⅝inch high) shall be prepared on a steel test plate in accordance with MIL-S-24238 using the SAW process with the flux to be tested in combination with a ⅛-inch diameter electrode (MIL-70S-1, -2, -3, -4, -5, and -6). Welding parameters shall be as follows:

Electrode stickout: 1 to 1½ inches

Welding current: 500±10 amperes, DCEP

Arc voltage: 28±1 volts

Travel speed: 15±1 inches per minute

Interpass temperature: 300 °F maximum

b. A second weld test pad shall be made using the same parameters except that the arc voltage shall be increased to 36±1 volts.

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c. The top surfaces of the weld pads shall be ground or machined to clean metal. Then samples for analysis shall be removed by machining. No weld metal shall be removed within ½ inch of the base plate. The sample analyzed shall be N and the absolute shall be of the base plate. The samples shall be chemically analyzed for silicon and manganese in accordance with ASTM E350. The N shall be calculated by the following Formula using the absolute values of the change in weight percentage of silicon and manganese levels:

$$N = 100[|\Delta Si| + |\Delta Mn|]$$

4.2.2 Alloy-cored electrode/flux combinations (MIL-80 series electrodes and fluxes). When qualification testing alloy-cored electrodes for the SAW process, each specific brand name electrode and flux combination shall be tested and listed as a unique combination on the QPL (see 6.2).

4.3 Conformance inspection. Conformance inspection shall include the examinations of MIL-DTL-23765 and the tests specified in table XI.

4.3.1 MIL-80 series bare, solid electrodes and rods.

4.3.1.1 Lot. For purposes of sampling and conformance inspection, a lot of electrodes or rods shall be the quantity of one size and type alloy produced in accordance with MIL-DTL-23765.

4.3.2 Certification of conformance. The certification of conformance shall be in accordance with MIL-DTL-23765.

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TABLE XI. Tests required for conformance inspection.

MIL-types [2/]	Form	Size	Tests required for conformance inspection [1/]									Test procedure	Requirements
			Cast [3/]	Helix [3/]	Chemical	RT	Tensile [4/]	Impact	Alloy identity	Welded test assembly	Diffusible hydrogen		
All MIL-70 series electrodes and rods	All	All	X	X	----	----	----	----	----	----	----	MIL-DTL-23765	MIL-DTL-23765
			----	----	X	----	----	----	----	----	----	3.4 herein and MIL-DTL-23765	Table I herein
			----	----	----	X	----	----	----	Table XII	----	MIL-DTL-23765	MIL-DTL-23765
			----	----	----	----	----	----	X	----	----	MIL-DTL-23765	MIL-DTL-23765
70S-1 70S-2 70S-3 70S-4 70S-5 70S-6	All	All	----	----	----	----	X	X	----	Table XII	----	MIL-DTL-23765	For the as-welded condition: Table IV herein When specified (see 6.2), for the stress-relieved condition [5/] Tables V, VI, as applicable, and XII (note 3) herein
70S-3H	All	All	----	----	----	----	X	X	----	Table XIII	----	MIL-DTL-23765	For the as-welded condition: Table IV herein
			----	----	----	----	----	----	----	----	X [6/]	3.7 herein	3.7 herein
70S-7 70S-8 70S-9	All, except form 6	All, except form 6	----	----	----	----	X	X	----	Table XII	----	MIL-DTL-23765	For the stress-relieved condition [5/]: Tables VI and XII (note 4) herein

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TABLE XI. Tests required for conformance inspection – Continued.

MIL-types [2/]	Form	Size	Tests required for conformance inspection [1/]									Test procedure	Requirements
			Cast [3/]	Helix [3/]	Chemical	RT	Tensile [4/]	Impact	Alloy identity	Welded test assembly	Diffusible hydrogen		
70S-F	N/A	N/A	----	----	----	X	----	----	----	Table XII or XIII, as applicable	[7/]	MIL-DTL-23765	MIL-DTL-23765
			----	----	----	----	X	X	----	Table XII or XIII, as applicable	----	MIL-DTL-23765	For the as-welded condition: Table IV herein When specified (see 6.2), for the stress-relieved condition [5/]: Table V and VI, as applicable, and Table XII (notes 3 and 4, as applicable) herein
All MIL-80 series electrodes and rods	All	All	X	X	----	----	----	----	----	----	----	MIL-DTL-23765	MIL-DTL-23765
			----	----	X	----	----	----	----	----	----	MIL-DTL-23765	Table II herein
			----	----	----	----	----	----	X	----	----	MIL-DTL-23765	MIL-DTL-23765
			----	----	----	X	----	----	----	Table XIV	----	MIL-DTL-23765	MIL-DTL-23765
			----	----	----	----	X	X	----	Table XIV	----	MIL-DTL-23765	For the stress-relieved condition [5/]: Tables VI and XIV (note 3) herein

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TABLE XI. Tests required for conformance inspection – Continued.

MIL-types [2/]	Form	Size	Tests required for conformance inspection [1/]									Test procedure	Requirements
			Cast [3/]	Helix [3/]	Chemical	RT	Tensile [4/]	Impact	Alloy identity	Welded test assembly	Diffusible hydrogen		
80S-1F 80S-2F	N/A	N/A	----	----	----	X	----	----	----	Table XIV	----	MIL-DTL-23765	MIL-DTL-23765
			----	----	----	----	X	X	----	Table XIV	----	MIL-DTL-23765	For the stress-relieved condition [5/]: Tables VI and XIV (note 3) herein

NOTES:

1/ The plastic spool burn test shall be in accordance with MIL-DTL-23765.

2/ Including applicable suffixes (see 3.3).

3/ Not applicable to form 6.

4/ Includes ultimate tensile strength, 0.2 percent offset yield strength, and percent elongation in 2 inches.

5/ MIL-70S-1, 70S-2, 70S-3, 70S-4, 70S-5, 70S-6, 70S-F, 80S-1, 80S-2, 80S-3, and 80S-F shall have the following suffixes added to the designation when tested in the stress-relieved condition:

SR1 – Stress-relieved in accordance with 3.5.1.1 (not applicable to MIL-80 series filler materials).

SR2 – Stress-relieved in accordance with 3.5.1.2, method (a).

SR3 – Stress-relieved in accordance with 3.5.1.2, method (b).

Examples as follows: MIL-70S-1SR1, MIL-70S-1FSR1.

6/ Diffusible hydrogen testing is required for SAW only.

7/ MIL-70S-3HF flux requires diffusible hydrogen testing in accordance with 3.7 herein.

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TABLE XII. Conformance requirements for test assemblies using MIL-70 series (except for MIL-70S-3H) bare, solid electrodes and rods, alloy-cored electrodes, and fluxes. [1/, 2/]

Purpose	Conformance		
Conditions (MIL-70S-1, -2, -3, -4, -5, and -6)	As-welded and, when specified (see 6.2), stress-relieved [3/]		
Conditions (MIL-70S-7, -8, and -9)	Unless otherwise specified (see note [4/]), stress relief shall be in accordance with 3.5.1 and 3.5.1.2 (method (b)).		
Welding parameters [5/, 6/]	----	----	----
Welding process (MIL-70S-1 -2, -3, -4, -5, and -6)	SAW	GMAW	GTAW (mechanized or automatic only) [7/, 8/]
Welding process (MIL-70S-7, -8, and -9)	SAW	----	----
Position	1G [7/]	1G [7/]	1G [7/]
Electrode size (inch)	3/32 and larger	less than 3/32	all sizes
Polarity	DCEP [8/]	DCEP [8/]	DCEN [9/]
Flux/shielding gas	70S-F [10/]	[11/]	Argon
Minimum preheat/maximum interpass temperature range (°F) [12/, 13/]	60/300±25	60/300±25	60/300±25
Number of weld beads	----	----	35 to 55
Joint configuration (as-welded and stress-relieved)	figure 1	figure 1	figure 1
Base material	[14/]	[14/]	[14/]
Thickness (T) (inch)	3/4	3/4	3/4
Width (W) (inches) (approximate)	10	10	10
Length (L) (inches) (minimum)	12	12	16
Root opening (R) (inch)	1/2	1/2	1/2
Root face (F) (inch) (maximum)	1/16	1/16	1/16
Included bevel angle (θ) (degrees)	45	45	45
Backing strap dimensions (t by w) (inches)	1/4 minimum by 1	1/4 minimum by 1	1/4 minimum by 1
Test specimens (number/type) (as-welded) [15/]	1 Tensile 5 CVN	1 Tensile 5 CVN	1 Tensile 5 CVN
Test specimens (number/type) (stress-relieved) [15/]	1 Tensile 5 CVN	1 Tensile 5 CVN	1 Tensile 5 CVN

NOTES:

1/ When a heat of metal is processed into electrodes and rods, weld tests shall be required only with electrodes.

2/ Including applicable suffixes (see 3.3).

3/ When specified (see 6.2), weldments shall be tested in the stress-relieved condition (in addition to the as-welded condition). The specified stress relief shall be in accordance with 3.5.1, 3.5.1.1, or 3.5.1.2 (either method (a) or (b)).

4/ Weldments shall be tested in the stress-relieved condition. When specified (see 6.2), stress relief shall be in accordance with 3.5.1 and 3.5.1.2 (method (a)).

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TABLE XII. Conformance requirements for test assemblies using MIL-70 series (except for MIL-70S-3H) bare, solid electrodes and rods, alloy-cored electrodes, and fluxes – Continued. 1/, 2/

NOTES (continued):

5/ The manufacturer's procedure shall reflect all essential elements necessary for welding the test assembly.

6/ Welding currents and pass sequence shall be in accordance with sound welding practices and as recommended by the manufacturer.

7/ 1G is the welding position (flat) as defined by AWS A3.0M/A3.0.

8/ Direct Current Electrode Positive (DCEP).

9/ Direct Current Electrode Negative (DCEN).

10/ 70S-F is the generic designator for flux that is qualified to this specification. The appropriate qualified flux shall be used with the MIL-type electrode being tested (e.g., 70S-1F flux with 70S-1 electrode or 70S-1F flux with 70S-1C alloy-cored electrode).

11/ Unless otherwise specified (see 6.2), the argon, oxygen, and carbon dioxide used for this test shall be in accordance with AWS A5.32M/A5.32. Shielding gas for the GMAW process shall be as follows:

a. MIL-70S-1: Argon plus 2 percent oxygen
b. MIL-70S-2, 70S-3, 70S-4, and 70S-6: Argon plus 2 percent oxygen and carbon dioxide, or argon plus 2 percent oxygen when specified (see 6.2) or only carbon dioxide when specified (see 6.2)
c. MIL-70S-5: Carbon dioxide

12/ The assembly shall be at room temperature at the start of welding, but not less than the required minimum preheat temperature. Continuous welding shall be employed in order to achieve the required interpass temperature. The interpass temperature range shall be maintained until welding is completed.

13/ After completion of each pass, the interpass temperature of the assembly shall be checked and the assembly shall be allowed to cool or shall be reheated, as required to maintain the assembly temperature in the required range prior to the resumption of welding. Any required cooling of test assemblies shall be done in still, room temperature air and assemblies shall not be force cooled with compressed air or by any other means.

14/ For each MIL-type, the base metal steel used for the tests required herein shall be in accordance with one of the following or equivalent, as specified (see 6.2):

a. Grade B steel of MIL-S-22698
b. Composition C of MIL-DTL-23194 or MIL-S-24238
c. Grade 70 of ASTM A516/A516M
d. Grade C of ASTM A285/A285M

15/ The tensile specimen shall be 0.500 inch in diameter in accordance with MIL-DTL-23765.

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TABLE XIII. Conformance requirements for test assemblies using MIL-70S-3H bare, solid electrodes and rods, alloy-cored electrodes, and flux. [1/, 2/]

Purpose	Conformance				
Condition (MIL-70S-3H)	As-welded only				
Welding parameters [3/, 4/]	----		----		----
Welding process (MIL-70S-3H)	SAW		GMAW		GTAW (mechanized or automatic only) [5/, 6/]
Position	1G [5/]		1G [5/]		1G [5/]
Electrode size (inch)	3/32 and larger		less than 3/32		1/16 and smaller
Polarity	DCEP [6/]		DCEP [6/]		DCEN [7/]
Flux/shielding gas	70S-3HF [8/]		[9/]		Argon
Cooling rate	High	Low	High	Low	-----
Heat input (kJ/in) [10/]	30+2/-5	75+2/-5	30+2/-5	75+2/-5	
Number of weld beads	----	----	----	----	35 to 55
Minimum preheat/maximum interpass temperature (°F) [11/]	60/150	275/300	60/150	275/300	60/300±25
Position	1G [5/]	1G [5/]	1G [5/]	3G [12/]	1G [5/]
Joint configuration (as-welded and stress-relieved)	figure 1		figure 1		figure 1
Base material	[13/]		[13/]		[13/]
Thickness (T) (inch)	3/4		3/4		3/4
Width (W) (inches) (approximate)	10		10		10
Length (L) (inches) (minimum)	12		12		12
Root opening (R) (inch)	1/2		1/2		1/2
Root face (F) (inch) (maximum)	1/16		1/16		1/16
Included bevel angle (θ) (degrees)	45		45		45
Backing strap dimensions (t by w) (inches)	1/4 minimum by 1		1/4 minimum by 1		1/4 minimum by 1
Test specimens (number/type) (as-welded) [14/]	1 Tensile 2 Bend 5 CVN		1 Tensile 2 Bend 5 CVN		1 Tensile 2 Bend 5 CVN
Test specimens (number/type) (stress-relieved) [14/]	1 Tensile 5 CVN		1 Tensile 5 CVN		1 Tensile 5 CVN

NOTES:

1/ When a heat of metal is processed into electrodes and rods, weld tests shall be required only with electrodes.

2/ Including applicable suffixes (see 3.3).

3/ The manufacturer's procedure shall reflect all essential elements necessary for welding the test assembly.

4/ Welding currents and pass sequence shall be in accordance with sound welding practices and as recommended by the manufacturer.

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TABLE XIII. Conformance requirements for test assemblies using MIL-70S-3H bare, solid electrodes and rods, alloy-cored electrodes, and flux – Continued. [1/, 2/]

NOTES (continued):

5/ 1G is the welding position (flat) as defined by AWS A3.0M/A3.0.

6/ Direct Current Electrode Positive (DCEP).

7/ Direct Current Electrode Negative (DCEN).

8/ MIL-70S-3HF qualified flux shall be used.

9/ Unless otherwise specified (see 6.2), the argon, oxygen, and carbon dioxide used for this test shall be in accordance with AWS A5.32M/A5.32. Shielding gas for the GMAW process shall be Argon plus 2 percent oxygen and carbon dioxide, or argon plus 2 percent oxygen when specified (see 6.2) or only carbon dioxide when specified (see 6.2)

10/ The listed heat input shall be the average heat input considering all-weld passes. The root layer may be excluded from all heat input calculations.

11/ After the joint has been tacked, the assembly shall be preheated as required. After completion of each pass, the interpass temperature of the assembly shall be checked and the assembly shall be allowed to cool or shall be reheated, as required to maintain the assembly temperature in the required range prior to the resumption of welding. Any required cooling of test assemblies shall be done in still, room temperature air and assemblies shall not be force cooled with compressed air or by any other means.

12/ 3G is the welding position (vertical) as defined by AWS A3.0M/A3.0.

13/ The base metal steel used for the test required herein shall be in accordance with ASTM A945/A945M, grade 65.

14/ The tensile specimen shall be 0.500 inch in diameter in accordance with MIL-DTL-23765.

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TABLE XIV. Conformance requirements for test assemblies using MIL-80 series bare, solid electrodes and rods, alloy-cored electrodes, and fluxes. 1/, 2/

Purpose	Conformance			
Condition	Stress-relieved 3/			
Welding parameters 4/, 5/	----	----	----	----
Position/electrode size (inch)	1G 6/ 3/32 through 1/4	3G 7/ 0.020 through 0.045	1G 6/ 0.052 through 5/64	1G 6/ 1/16 and smaller
Welding process (MIL-80S-1 and -2)	SAW	----	----	----
Welding process (MIL-80S-3)	----	GMAW pulsed	GMAW spray	GTAW 8/, 9/
Polarity	DCEP 8/	DCEP 8/	DCEP 8/	DCEN 9/
Flux/shielding gas 10/	80S-F 11/	Argon plus 2 percent oxygen or argon plus 5 percent CO_2, as specified (see 6.2)	Argon plus 2 percent oxygen or argon plus 5 percent CO_2, as specified (see 6.2)	Argon
Minimum preheat/maximum interpass temperature (°F) 12/	250/500	250/500	250/500	250/500
Joint configuration	figure 1	figure 1	figure 1	figure 1
Base material	13/	13/	13/	13/
Thickness (T) (inch)	3/4	3/4	3/4	3/4
Width (W) (inches) (minimum)	10	10	10	10
Length (L) (inches) (minimum)	12	12	12	12
Root opening (R) (inch)	1/2	1/2	1/2	1/2
Root face (F) (inch)	0 to 3/16	0 to 3/16	0 to 3/16	0 to 3/16
Included bevel angle (θ) (degrees)	45	45	45	45
Backing strap dimensions (t by w) (inches)	1/4 to 3/8 thick by 1	1/4 to 3/8 thick by 1	1/4 to 3/8 thick by 1	1/4 to 3/8 thick by 1
Test specimens (number/type) 14/	1 Tensile 5 CVN	1 Tensile 5 CVN	1 Tensile 5 CVN	1 Tensile 5 CVN

NOTES:

1/ When a heat of metal is processed into electrodes and rods, weld tests shall be required only with electrodes.

2/ Including applicable suffixes (see 3.3).

3/ Weldments shall be tested in the stress-relieved condition. The stress relief shall be in accordance with 3.5.1 and 3.5.1.2. Holding time shall be in accordance with 3.5.1.2, method (a) or (b), as specified (see 6.2). After completion of the weld and stress relief, it shall be allowed to cool, the weld reinforcement and backing strip shall be removed flush with the base plate. The backing strip may be removed by the air carbon arc gouging process. No base metal shall be removed within 1/2 inch of the edges of the face of the weld by flame cutting. Only sawing, machining, or both shall be used.

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TABLE XIV. Conformance requirements for test assemblies using MIL-80 series bare, solid electrodes and rods, alloy-cored electrodes, and fluxes – Continued. 1/, 2/

NOTES (continued):

4/ The manufacturer's procedure shall reflect all essential elements necessary for welding the test assembly.

5/ Welding currents and pass sequence shall be in accordance with sound welding practices and as recommended by the manufacturer. Any required cooling of test assemblies shall be done in still, room temperature air and prior to resumption of welding, the assembly shall be reheated to the minimum preheat temperature.

6/ 1G is the welding position (flat) as defined by AWS A3.0M/A3.0.

7/ 3G is the welding position (vertical) as defined by AWS A3.0M/A3.0.

8/ Direct Current Electrode Positive (DCEP).

9/ Direct Current Electrode Negative (DCEN).

10/ Shielding gases shall be in accordance with AWS A5.32M/A5.32.

11/ 80S-F is the generic designator for flux that is qualified to this specification. The appropriate qualified flux shall be used with the MIL-type electrode being qualified (e.g., 80S-1F flux with 80S-1 electrode or 80S-1F flux with 80S-1C electrode). For conformance testing of alloy-cored electrodes with the SAW process, the same brand name electrode and flux combination shall be used as qualified and listed on the QPL (see 4.2.2 and 6.2).

12/ After the joint has been tacked, the assembly shall be preheated. After completion of each pass, the interpass temperature of the assembly shall be checked and the assembly shall be allowed to cool or shall be reheated, as required to maintain the assembly temperature in the required temperature range prior to resumption of welding. Any required cooling of test assemblies shall be done in still, room temperature air and assemblies shall not be force cooled with compressed air or by any other means.

13/ Unless otherwise specified (see 6.2), base metal steel used for the tests required herein shall be in accordance with one of the following:

a. Ni-Cr-Mo steel in accordance with MIL-DTL-23194, composition A
b. ASTM A508/A508M, grade 2, class 1 or equivalent
c. Mn-Ni-Mo steel in accordance with MIL-S-24238, composition A
d. ASTM A302/A302M
e. ASTM A533/A533M, type B or C, class 1, or equivalent

14/ The tensile specimen shall be 0.500 inch in diameter in accordance with MIL-DTL-23765.

4.4 Inspection of packaging and marking. Inspection of packaging and marking shall be in accordance with MIL-DTL-23765.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. This specification covers the welding materials described in 6.1.1 through 6.1.15 that, when deposited using the specified welding processes (SAW, GMAW, and GTAW) on the applicable base metals specified herein, will meet the mechanical properties specified herein in the as-welded condition and the stress-relieved condition. Current stock of electrode types specified in this specification may be used until depleted; future requisitions for electrode types specified in this specification should refer to the current version of this specification.

6.1.1 MIL-70S-1. MIL-70S-1 electrodes are primarily used for multiple pass welding killed steel.

6.1.2 MIL-70S-2. MIL-70S-2 electrodes contain aluminum, zirconium, and titanium in addition to the silicon and manganese content. These electrodes are particularly adapted to producing sound welds in semi-killed and rimmed steels as well as killed steels.

6.1.3 MIL-70S-3. MIL-70S-3 electrodes have a higher silicon content than MIL-70S-1 electrodes. These electrodes can be used to make multiple pass welds in semi-killed and killed steels.

6.1.4 MIL-70S-3H. MIL-70S-3H electrodes and fluxes meet requirements for welding Navy HSLA-65 steel.

6.1.5 MIL-70S-4. MIL-70S-4 electrodes have a higher silicon content than MIL-70S-3 electrodes. These electrodes are used where welding application requires a higher arc voltage or conditions requiring more deoxidization than MIL-70S-3 electrodes.

6.1.6 MIL-70S-5. MIL-70S-5 electrodes contain manganese, silicon, and aluminum. These electrodes should be used only for structural work when exceptional deoxidization of weld metal is required.

6.1.7 MIL-70S-6. MIL-70S-6 electrodes contain a high manganese and silicon content and produces welds with high impact properties.

6.1.8 MIL-70S-7. MIL-70S-7 electrodes contain a high manganese mild steel composition that is capable of attaining properties specified herein when deposited by the SAW process.

6.1.9 MIL-70S-8. MIL-70S-8 electrodes are an intermediate manganese-molybdenum composition that is used in SAW of mild steel.

6.1.10 MIL-70S-9. MIL-70S-9 electrodes are high manganese-molybdenum compositions for SAW that retains the highest mechanical properties after stress relief.

6.1.11 MIL-80S-1, MIL-80S-2, and MIL-80S-3. MIL-80S-1 and MIL-80S-2 electrodes are suitable for welding MIL-DTL-23194, composition A steels; MIL-S-24238, composition A steels; ASTM A508/A508M, grade 2, class 1 steels; and ASTM A533/A533M and ASTM A302/A302M steels using the SAW process. MIL-80S-3 is suitable for welding the same steels using the GMAW and GTAW processes.

6.1.12 Special MIL-type with suffix “RC”.

6.1.12.1 MIL-70 series. MIL-70 series electrodes and rods with the suffix “RC” have restricted copper and phosphorus content.

6.1.12.2 MIL-80 series. MIL-80 series electrodes and rods with the suffix “RC” have restricted copper content.

6.1.13 MIL-type with suffix “C”. Suffix “C” indicates alloy-cored electrodes intended for use with the GMAW and SAW processes only.

6.1.14 MIL-70S-F. MIL-70S-F series fluxes are neutral granular fluxes intended for SAW with MIL-70 series electrodes.

6.1.15 MIL-80S-F. MIL-80S-F series fluxes are neutral granular fluxes intended for SAW with MIL-80 series electrodes.

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6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of the specification.

b. For MIL-70 series electrodes and rods, whether the copper and phosphorus content should be restricted (see note 2/ of table I and 6.1.12.1).

c. For MIL-80 series electrodes and rods, whether the copper content should be restricted (see note 5/ of table II and 6.1.12.2).

d. Mechanical properties of as-welded weld metal if other than as specified in 3.5. (see 3.5).

e. For MIL-70S-1, -2, -3, -4, -5, and -6, whether the tests must be conducted in the stress-relieved condition in accordance with 3.5.1, 3.5.1.1, and 3.5.1.2 (either method (a) or (b)) (see table XI and note 3/ of table XII).

f. For electrodes MIL-70S-7, -8, and -9, whether the tests must be conducted in the stress-relieved condition in accordance with 3.5.1 and 3.5.1.2 (method (a)) (see note 4/ of table XII).

g. Other base metal steel to be used in tests, if required (see note 14/ of table XII or note 13/ of table XIV, as applicable).

h. For electrodes MIL-70S-2, -3, -4, -6 and 70S-3H, whether only the argon plus 2 percent oxygen weld test or only the carbon dioxide weld test is required (see note 11/ of table XII and note 9/ of table XIII.).

i. For MIL-80S-3, whether the GMAW shielding gas should be argon plus 2 percent oxygen or argon plus 5 percent CO_2 (see table XIV).

j. For MIL-80S-1, -2, and -3, whether the holding time at temperature for the stress relief treatment should be in accordance with 3.5.1.2 (either method (a) or (b)) (see note 3/ of table XIV).

k. For MIL-80S-1 and -2, which specific brand name electrode and flux combination(s) must be used for conformance testing (see note 9/ of table XIV and 4.2.2). See current QPL for qualified combinations.

l. Packaging requirements (see 5.1).

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Products List QPL No. 23765 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard DC 20376-5160 or emailed to CommandStandards@navy.mil. An online listing of products qualified to this specification may be found in the Qualified Products Database (QPD) at https://assist.dla.mil.

6.4 Subject term (key word) listing.

Flux

Gas metal-arc welding (GMAW)

Gas tungsten-arc welding (GTAW)

Submerged-arc welding (SAW)

70S-1

70S-2

70S-3

70S-3H

70S-4

70S-5

70S-6

70S-7

70S-8

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70S-9

70S-F

80S-1

80S-2

80S-3

80S-F

6.5 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

CONCLUDING MATERIAL

Custodian:
Navy – SH

Preparing activity:
Navy − SH
(Project 3439-2022-002)

Review activity:
DLA – GS

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